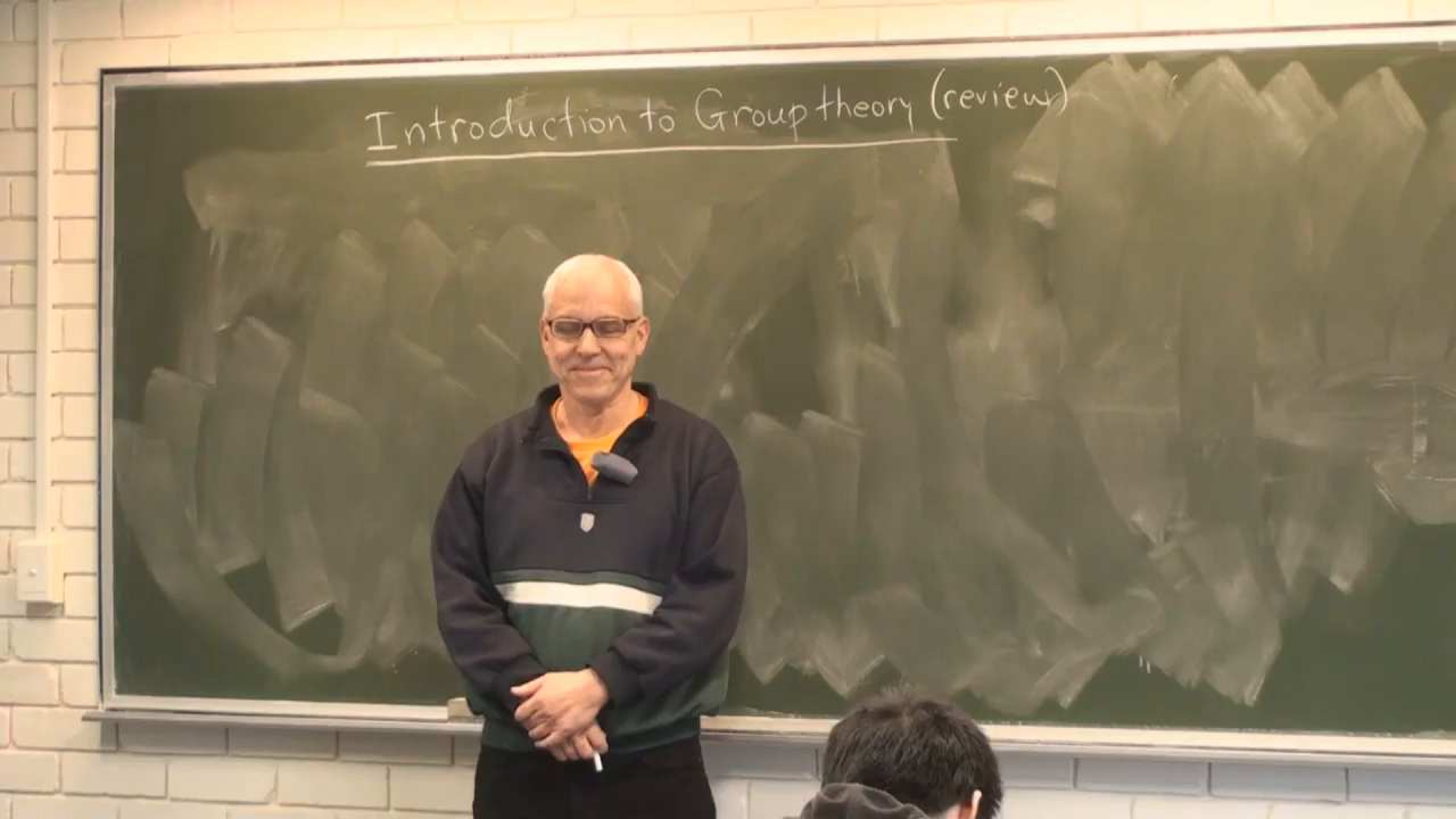
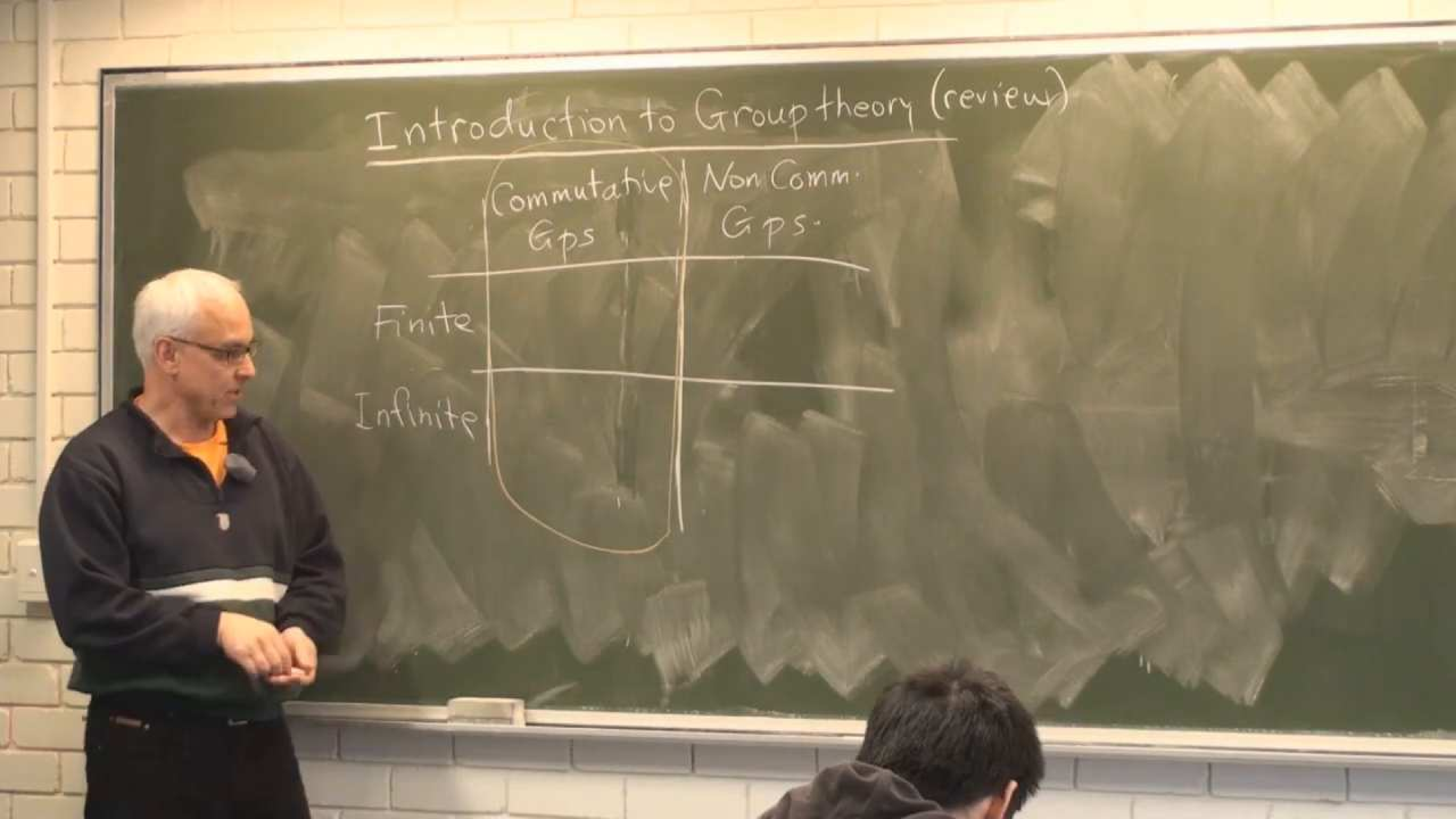
Algebraic Topology: a beginner's course

This lecture:

AlgTopReview2: Introduction to group theory


Introduction to Group theory (review)


Introduction to Group theory (review)
Commutative Gps
Non Comm. Gps.
Finite
Infinite

3) **Identity** There is a special element $e \in G$ such that

$$e * a = a * e = a$$

for any $a \in G$

4) **Inverse** For every $a \in G$ there is a $b \in G$ such that

$$a * b = b * a = e$$

We say b is the inverse of a. $\boxed{b = a^{-1}}$

Ex $G = \{0, 1, 2, 3, 4\}$

$a * b \equiv a + b \pmod 5$

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	$\boxed{4}$	0
2	2	3	4	0	1
3	3	$\boxed{4}$	0	1	2
4	4	0	1	2	3

$1+3$

$3+1$

$(2+3)+4 = 0+4 = 4$

$2+(3+4) = 2+2 = 4$

Convention. write $* = +$ (for comm. gps only!)

Ex. 2

$+$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

$G = \{0, 1, 2\} = \mathbb{Z}_3$.

Prev. ex. $\mathbb{Z}_5$

In general $\mathbb{Z}_n$ for $n = 1, 2, 3, \ldots$

$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$

$+$ = addition (mod n)

...ntion. write $* = +$ (for comm. gps only!)

	0	1	2
	0	1	2
	1	2	0
	2	0	1

$\{0,1,2\} = \mathbb{Z}_3$

Ex. $\mathbb{Z}_5$

In general $\mathbb{Z}_n$ for $n = 1, 2, 3, \ldots$

$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$

$+ =$ addition $(\mathrm{mod}\ n)$

Ex 3 $G = \mathbb{Z}$ operation $+$

Identity $= 0$ $\qquad 5 + (-5) = 0$

Inverse of a is $-a$.

Multiples. $a \in \mathbb{Z}$. $a + a \equiv 2a$ $\quad a + a + a \equiv 3a$ etc.

x operation + identity = 0

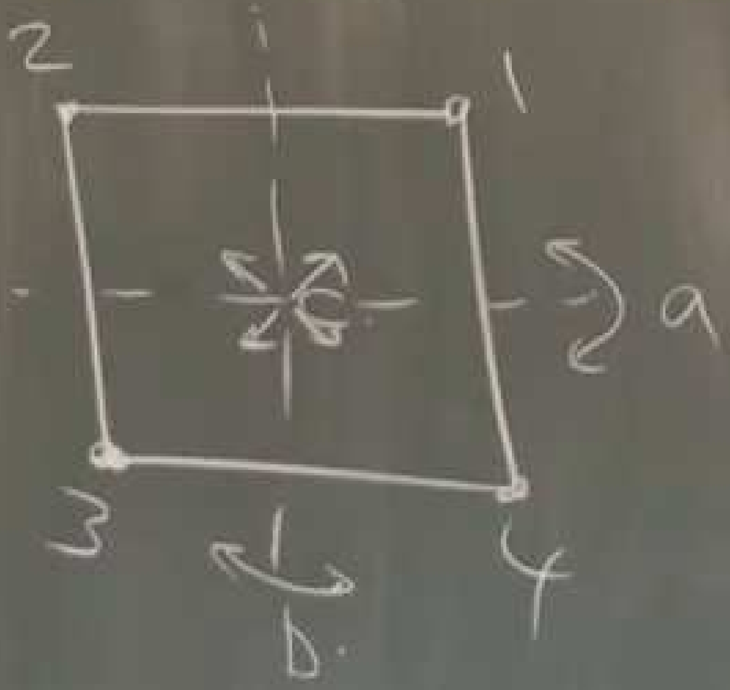
inverse $a = -a$

(...tiplicative notation)

$\times$	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

group

$G = \{1, a, b, c\}$



$a: 1 \leftrightarrow 4,\ 2 \leftrightarrow 3$ $\quad (14)(23)$

$b: 1 \leftrightarrow 2,\ 3 \leftrightarrow 4$ $\quad (12)(34)$

$c: 1 \leftrightarrow 3,\ 2 \leftrightarrow 4.$ $\quad (13)(24)$

$*$ = composition of transformations

$b \circ a: 1 \xrightarrow{a} 4 \xrightarrow{b} 3$

$2 \to 3 \to 4$

$3 \to 2 \to 1$

$4 \to 1 \to 2$

a, b, c are permutations of $\{1, 2, 3, 4\}$.

$\boxed{b \circ a = c}$

Ex. $G = \mathbb{Z}$; $0, \pm 1, \pm 2, \pm 3, \ldots$

H: even integers ← a subgroup.

$$H = 2\mathbb{Z}$$

multiples of 3 : $3\mathbb{Z}$: $\ldots, -6, -3, 0, 3, 6, \ldots$

also a subgroup.

Every subgroup of $\mathbb{Z}$ is of the form

$n\mathbb{Z}$ for $n = 0, 1, 2, 3, 4, \ldots$

$\mathbb{Z}$: $0, \pm 1, \pm 2, \pm 3, \dots$

even integers $\leftarrow$ a subgroup.

$H = 2\mathbb{Z}$

les of 3 : $3\mathbb{Z}$ $\dots, -6, -3, 0, 3, 6, \dots$

also a subgroup.

ubgroup of $\mathbb{Z}$ is of the form

or $n = 0, 1, 2, 3, 4, \dots$

Direct sums of gr

Ex $G = \mathbb{Z}$: $0, \pm 1, \pm 2, \pm 3, \ldots$

H : even integers ← a subgroup

$H = 2\mathbb{Z}$

multiples of 3 : $3\mathbb{Z}$ $\ldots -6, -3, 0, 3, 6, \ldots$

also a subgroup

Every subgroup of $\mathbb{Z}$ is of the form

$n\mathbb{Z}$ for $n = 0, 1, 2, 3, 4, \ldots$

Direct sums of groups:

Let G, H be groups (comm) w/ operations $+_G$, $+_H$

$G \oplus H$: (g, h) $g \in G$, $h \in$

$(g_1, h_1) + (g_2, h_2) \equiv (g_1 + g_2, h_1 + h_2)$

$0_{G \oplus H} = (0_G, 0_H)$ identity

The typical cyclic groups are $\mathbb{Z}_n$

$$\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$$

[1 acts as a generator, there are others too ($(k, n) = 1 \Rightarrow k$ is a generator)]

Fund thm of finite comm groups

Every finite comm gp is isomorphic to a sum of cyclic groups $G \cong \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \cdots \oplus \mathbb{Z}_{n_k}$

(for some n_i, ... n_i [illegible])

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also see:

AlgTopReview3: More on commutative groups